

Supplementary Information

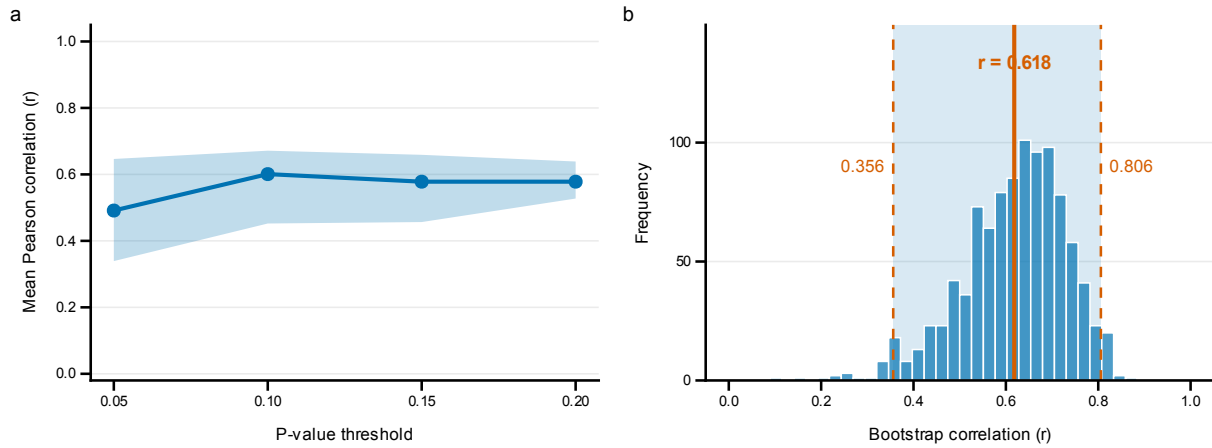
A Calibrated Transcriptomic Atlas of Porcine Development Reveals Conserved Molecular Programs with Humans

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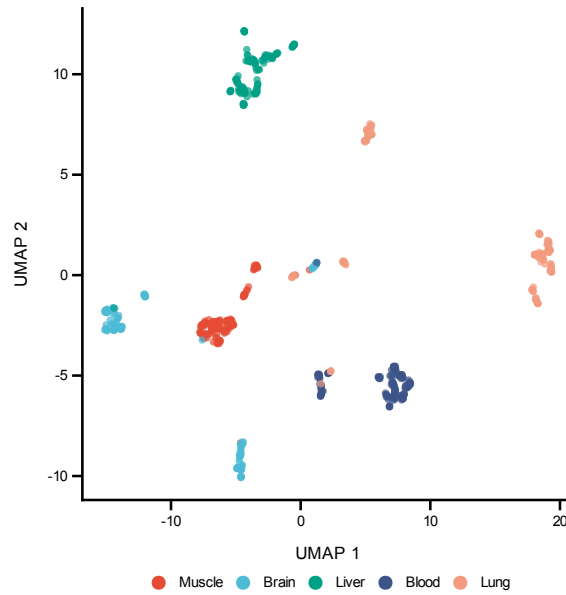
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Supplementary Figures

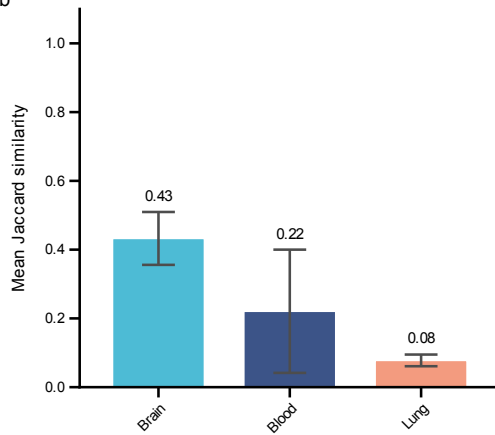


Supplementary Figure 1: Cross-species sensitivity analysis of conserved developmental markers. a, Mean Pearson correlation between pig and human developmental fold changes across significance thresholds ranging from $p < 0.01$ to $p < 0.20$. Shaded region indicates the range (minimum–maximum) across different gene set sizes ($n = 15$ – 50 genes). **b**, Bootstrap distribution of correlation coefficients ($n = 1,000$ iterations) for the optimal parameter combination ($p < 0.10$, $n = 36$ genes). Solid vertical line indicates the observed correlation ($r = 0.618$); dashed lines denote the 95% bootstrap confidence interval boundaries (0.36–0.81).

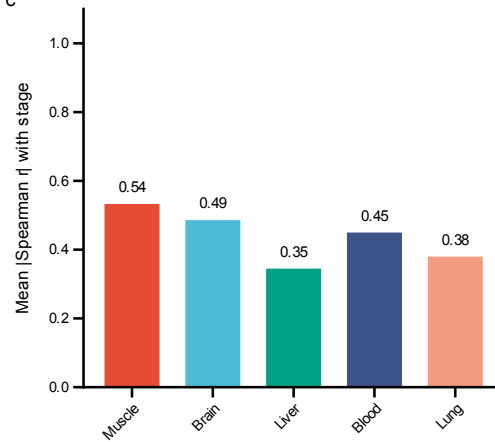
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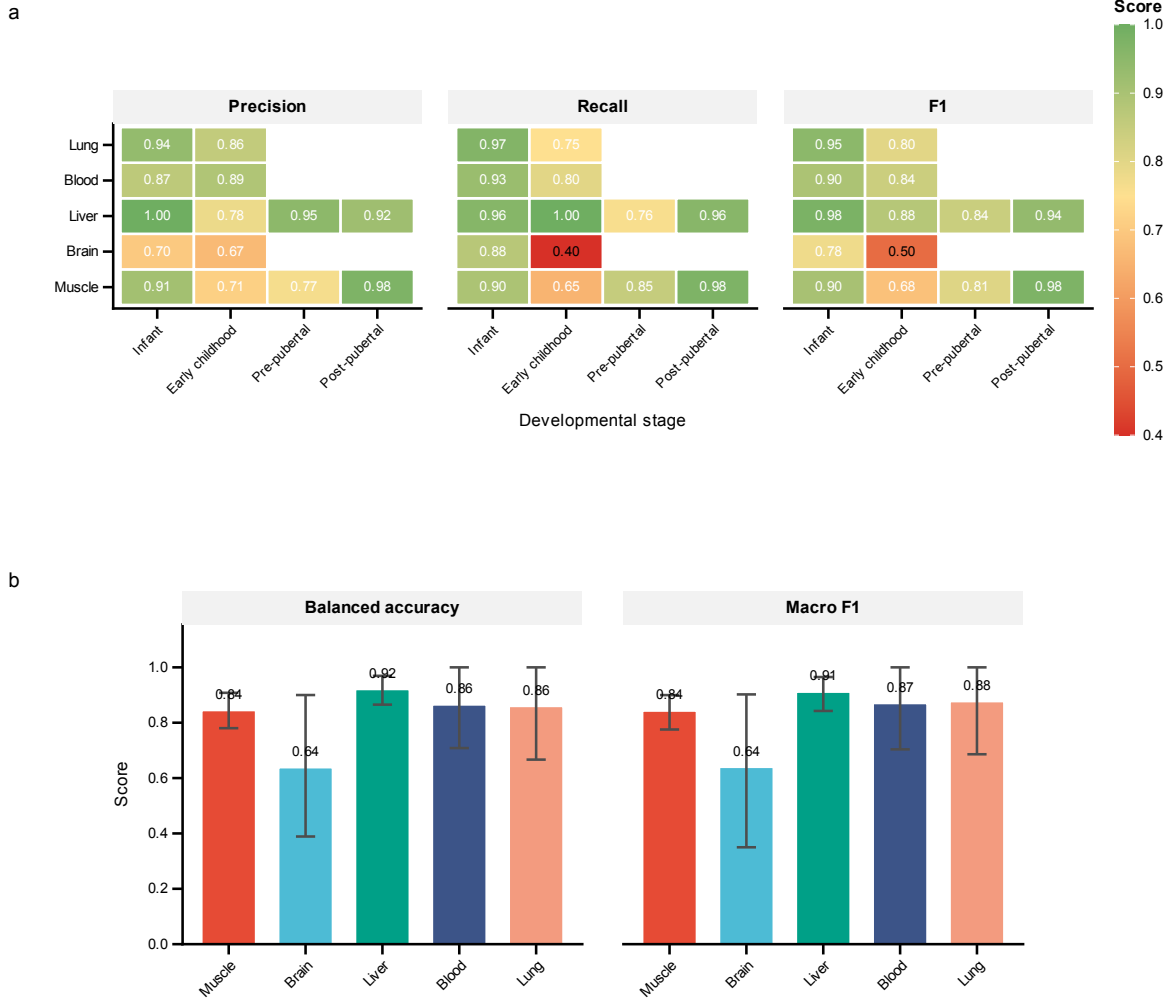
b



c



Supplementary Figure 2: Tissue specificity and feature selection stability analysis. **a**, Uniform manifold approximation and projection (UMAP) of gene expression profiles across five tissues (muscle, brain, liver, blood, lung), demonstrating distinct tissue-specific clustering patterns ($n = 150$ samples per tissue, balanced sampling; top 500 variable genes). **b**, Within-tissue feature selection stability quantified as mean Jaccard similarity index of the top 50 selected features across five independent random seeds (brain, blood, lung; stability analysis not performed for muscle and liver). Error bars represent standard deviation; higher values indicate more reproducible feature selection. **c**, Mean absolute Spearman correlation between top-ranked features and developmental stage for each tissue.



Supplementary Figure 3: Extended classification performance metrics across tissues and developmental stages. **a**, Heatmap of per-class precision, recall, and F1 scores for each developmental stage across all five tissues. Colour scale ranges from 0.4 (red, poor performance) to 1.0 (green, excellent performance). Missing values indicate developmental stages not present in that tissue's classification scheme. **b**, Overall performance summary showing balanced accuracy and macro-averaged F1 scores evaluated on the held-out test set (30% of samples) for each tissue. Error bars indicate 95% bootstrap confidence intervals from 1,000 resampling iterations.

Supplementary Tables

Table 1: Train versus test performance comparison. Test set performance (30% held out) is reported for all tissues. Train set metrics are not shown as models were evaluated only on held-out data to prevent overfitting.

Tissue	Scheme	n	Balanced Acc.	F1-macro	F1-weighted	MAE
Muscle	4-class	761	0.844	0.842	0.908	0.10
Brain	2-class	43	0.637	0.639	0.671	–
Liver	4-class	309	0.920	0.910	0.913	0.09
Blood	2-class	78	0.864	0.869	0.874	–
Lung	2-class	129	0.859	0.876	0.921	–

Table 2: Per-class performance metrics by tissue. Precision, recall, and F1 scores for each developmental stage within each tissue. n indicates the number of test samples per class.

Tissue	Stage	n	Precision	Recall	F1
Muscle	Infant	58	0.912	0.897	0.904
Muscle	Early childhood	23	0.714	0.652	0.682
Muscle	Pre-pubertal	27	0.767	0.852	0.807
Muscle	Post-pubertal	121	0.975	0.975	0.975
Brain	Infant	8	0.700	0.875	0.778
Brain	Early childhood	5	0.667	0.400	0.500
Liver	Infant	26	1.000	0.962	0.980
Liver	Early childhood	18	0.783	1.000	0.878
Liver	Pre-pubertal	25	0.950	0.760	0.844
Liver	Post-pubertal	24	0.920	0.958	0.939
Blood	Infant	14	0.867	0.929	0.897
Blood	Early childhood	10	0.889	0.800	0.842
Lung	Infant	31	0.938	0.968	0.952
Lung	Early childhood	8	0.857	0.750	0.800

Table 3: Functional annotations of key developmental markers. Annotations compiled from UniProt, GeneCards, and literature review. All listed genes show conserved developmental expression patterns between pig and human muscle tissue. FC = $\log_2$ fold change (adult/infant); positive values indicate higher expression in adults (upregulated during development), negative values indicate higher expression in infants (downregulated during development). Pig FC derived from pig GTEx data (GEO: GSE257558); Human FC from Schaiter *et al. Sci. Rep.* **14**, 22953 (2024).

Gene	Developmental Function	Pig FC	Human FC
<i>SORCS2</i>	Serves as a molecular switch directing trafficking of key signalling receptors to regulate the balance between neuronal growth and pruning, ensuring precise wiring of the developing nervous system.	−1.84	−0.95
<i>FGFRL1</i>	Acts as a “decoy” receptor dampening excessive growth factor signalling to ensure proper formation of critical structures including the diaphragm, kidneys, and skeleton.	−1.68	−2.10
<i>KREMEN1</i>	Functions as a critical negative regulator of Wnt signalling, partnering with Dkk1 to control cell survival and ensure proper patterning of organs including thymus, teeth, and neural tissues.	−1.25	−1.43
<i>ACHE</i>	Functions not only as an enzyme regulating chemical signalling to prevent cell damage but also as a structural adhesion molecule guiding physical growth of neurons and brain wiring.	−1.81	−0.53
<i>MSS51</i>	Acts as a skeletal muscle-specific regulator of metabolism, modulating mitochondrial respiration and energy expenditure. Highly expressed specifically in fast-twitch (glycolytic) muscle fibres; expression increases postnatally as fast-twitch fibres mature. Expression in slow-twitch (oxidative) fibres is reduced.	3.02	1.55
<i>DLK1</i>	Crucial regulator of the myogenic program that inhibits myoblast proliferation while promoting differentiation; shows high infant expression that declines as the muscle fiber population matures.	−1.37	−2.88
<i>ACTC1</i>	Cardiac α -actin expressed in fetal skeletal muscle; switches to skeletal actin (ACTA1) postnatally as a hallmark of muscle fiber maturation. LMOD2-ACTC1 interaction regulates myogenic differentiation.	−2.54	−0.28
<i>PDLIM3</i>	Acts as a structural Z-disc protein stabilising actin filaments at the muscle Z-disc, playing a critical role in ensuring structural integrity of striated muscle. Expression increases from Infant (low) through development to Adult stages (high), reflecting establishment of the mature contractile apparatus.	2.19	0.56

Table 4: Stage-by-stage expression trajectories of key developmental markers. Mean TPM values across porcine developmental stages for genes with conserved pig-human developmental patterns. Stage boundaries: Infant (0–20 d), Early childhood (21–59 d), Pre-pubertal (60–149 d), Post-pubertal (150–365 d), Adult (>365 d). “Peak Change” indicates the developmental transition with the largest fold change.

Gene	Infant	Early	Pre-pub	Post-pub	Adult	Peak Change
<i>SORCS2</i>	14.6	10.9	6.5	0.6	4.1	Pre-pub→Post-pub
<i>FGFRL1</i>	71.2	20.6	27.0	4.3	22.2	Infant→Early
<i>ACHE</i>	16.2	9.9	9.1	1.3	4.6	Pre-pub→Post-pub
<i>KREMEN1</i>	28.9	15.1	12.2	5.2	12.1	Pre-pub→Post-pub
<i>DLK1</i>	301.5	25.8	121.8	0.7	116.5	Infant→Early
<i>ACTC1</i>	9112.7	696.0	1125.4	57.6	1566.8	Pre-pub→Post-pub
<i>MSS51</i>	28.9	177.3	191.4	229.2	234.8	Infant→Early
<i>PDLIM3</i>	278.6	771.0	535.0	1221.2	1269.4	Infant→Early